Evidence Report

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Title: Workstation Image Examination III

Case: 26-T005

Date: 05/01/2026

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Document Revision History

Name	Revision Date	Version	Description
Tytus Gonzalez	2026-04-08	0.1	Draft
Tytus Gonzalez	2026-05-01	1.0	Final

Executive Summary

On 4/3/2026, a corporate workstation was reported to have had sensitive documents stolen and posted on a competitor's website. A forensic investigation was conducted using a forensic image of the workstation to determine the partition structure, verify the integrity of the disk, and identify relevant user and file activity associated with the incident. I, Tytus Gonzalez, prepared a Windows forensic virtual machine and conducted analysis using both WinHex and The Sleuth Kit in a controlled environment.

The investigation began with verifying the integrity of the disk image using hashing tools to ensure no corruption occurred during acquisition. The image was then opened in WinHex to perform a low-level analysis of the disk, beginning at offset 0x000. The first 440 bytes were examined to confirm the presence of boot code, and the disk signature located at offset 0x01B8 was recorded. The partition table at offset 0x01BE was analyzed and found to contain four entries, including unused entries. Partition 1 was identified as bootable and of type NTFS. The LBA start of the partition was calculated as 63. The partition size was determined to be approximately 10.7 GB and closely matches the total size of the disk image. Examination of the partition start revealed the NTFS boot sector signature, confirming the file system type and integrity. Based on these findings, the disk image was determined to use a Master Boot Record (MBR) partitioning scheme.

Following the initial disk analysis, further examination was conducted using The Sleuth Kit to analyze the NTFS file system and user activity. The partition offset of 63 sectors was used to perform file system analysis. The volume serial number was identified, and key file system parameters, including sector size (512 bytes) and cluster size (4096 bytes), were confirmed. The Master File Table (MFT) was located at cluster 786432, providing the foundation for file system metadata analysis.

User activity was examined by identifying interactive user profiles within the "Documents and Settings" directory. Three user accounts were identified: Administrator, Devon, and Jean. Further examination of user directories showed that Jean's Desktop contained two files, including a Microsoft Excel document (m57biz.xls), while Devon's Desktop contained no files.

Additional analysis was performed to verify whether the file m57biz.xls, which was identified on a competitor's website, existed on the system and contained matching content. Using Sleuth Kit tools, the file was located, its metadata reference identified, and its contents recovered. Analysis confirmed that the file exists on the disk image and that its contents are consistent with the exfiltrated data observed on the competitor's site. Metadata analysis revealed that the file was authored by Alison Smith and last saved by Jean, indicating user interaction prior to exfiltration.

Further examination of the file's NTFS attributes, including $STANDARD_INFORMATION, $FILE_NAME, and $DATA, provided insight into file size, allocation, and storage behavior. A forensic timeline was constructed using file system metadata to analyze the file's activity. The timestamps indicate that the file was created, accessed, and modified on 2008-07-19, with a subsequent MFT change occurring shortly thereafter. These findings provide a clear sequence of events associated with the file's existence and usage on the system.

This investigation demonstrates a comprehensive forensic analysis process, including verification of disk integrity, examination of partition structures, detailed file system analysis, file recovery, metadata interpretation, and timeline reconstruction. The combined use of low-level disk analysis and file system forensics provides a clear understanding of the structure, user activity, and potential relevance of data present on the workstation in relation to the reported data exfiltration incident.

Synopsis

The Professor provided the following questions.

Review the first 446 bytes

> **1. Does this image have boot code in the first 440 bytes?**
>
> **2. Give your observations including any plaintext in the code.**
>
> **3. What is the "disk signature" starting at offset 0x01B8? Provide the value as a hexadecimal value like 0x01234567; Give the correct value with endianness in consideration**

Review the partition table beginning at offset 0x01BE

> **1. How many partition entries are in the table? Include empty ones.**
>
> **2. Which partition entry is marked bootable?**
>
> **3. Partition entry 1 - What is the partition entry type? Give the hexadecimal value and the type (e.g. Linux, FAT32)**
>
> **4. Partition entry 1 - What is the LBA start address? Give the hexadecimal value with endianness in consideration**
>
> **5. Calculate the offset of the LBA start address. Sector size is 512 bytes or 0x200 in hexadecimal.**
>
> **6. Partition entry 1 - What is the partition size? Give the hexadecimal value with endianness in consideration**
>
> **7. Calculate the size of the partition in bytes. Sector size is 512 bytes or 0x200 in hexadecimal Size must be in decimal**
>
> **8. What is the difference, in bytes, between the actual size of the evidence file versus the value above?**

Navigate to Partition Entry One LBA Start Address Offset

> **1. What are the first 8 bytes at this offset? DO NOT consider endianness in this value**
>
> **2. Does this value support your partition type answer above?**
>
> **3. Give your observations including any plaintext in the code.**
>
> **4. Based on your observations, does this disk image use an MBR or GPT partitioning scheme? Explain why.**

1. Using ‘mmls’ , what is the starting sector of the NTFS/exFAT partition?

2. Using ‘fsstat’ , what is the volume serial number?

3. What is the sector size?

4. What is the cluster size?

5. How many sectors fit into one cluster?

6. What is the first cluster of the MFT?

7. Using ‘fls’ , what is the metadata reference (inode) for the folder “Documents and Settings”

8. Using this metadata reference and ‘fls’ , there are three directories representing interactive users (users that have logged into the system). What are the directory (user) names? Do not include “All Users”, “Default User”, “Local Service”, or “Network Service”.

9. Using ‘istat’ and the metadata reference for the folder “Devon”, review the $STANDARD_INFORMATION block. What is the create date timestamp?

10. Do the same for the folder “Jean”. What is the create date timestamp?

11. Based on this information, what user profile was created first?

12. Using ‘fls’ , review the contents of Jean’s “Desktop” folder. What are the file names of the two files?

13. Using ‘istat’ , what is the created date/timestamp of the file WITHOUT the “.url” extension?

14. Are there any files in Devon’s “Desktop” folder? If so, what are the file names?

15. Use ‘icat’ to recover the file from question 13 to disk. Run the ‘file’ command on the recovered file. According to the file output, who was the “author” and “last saved” by entries?

Examination 3

1. **Does the target file m57biz.xls exist on this system? Does it have the same content as the exfiled content?**
2. **Using ffind, what directory does the target file reside in?**
3. **What is the metadata reference for the target file?**
4. **What is the MFT entry for this file. (Hint: its the first number in the fls metadata reference)**
5. **Based on this information, what is the offset in the MFT for this entry. (Hint: this number should be in the 30 million)**
6. **Review the MFT for the target file, what is the length of $STANDARD_INFO in bytes?**
7. **What is the length of $FILE_NAME in bytes?**
8. **Is there another attribute after $FILE_NAME and before $DATA? What is it?**
9. **What is the length of $DATA in bytes?**
10. **What is the allocated size for this file?**
11. **What is the actual size for this file?**
12. **What is the file slack for this file?**
13. **Using fls, what is the metadata reference for the folder where this file resides?**
14. **Generate a bodyfile for this metadata reference only. How many entries are their?**
15. **Generate a timeline using mactime for this bodyfile. What is the modified, accessed, created (born) date timestamp for the targetfile? (Hint: ma.b)**
16. **What is the MFT changed date timestamp for the target file? (Hint: ..c.)**

Evidence Collected

The section outlines all physical evidence items collected

Name	Identifier (Tag)	Description
windows-image-01.dd	windows-image-01	DD image of Windows machine provided by instructor

This section provides details of the digital evidence collected

Name	windows-image-01.dd
Type:	DOS/MBR boot sector MS-MBR XP english at offset 0x12c "Invalid partition table" at offset 0x144 "Error loading operating system" at offset 0x163 "Missing operating system", disk signature 0x39bf39be; partition 1 : ID=0x7, active, start-CHS (0x0,1,1), end-CHS (0x3ff,254,63), startsector 63, 20948697 sectors
Size:	10737418240 bytes
MD5:	78a52b5bac78f4e711607707ac0e3f93
SHA1:	ba7dc57e08bb6e3393aee15c713ae04feadcd181
SHA256	4d4fc46f284630a69980340ee36d3ed486de424a9f456eedba0f7194cc7a991b

Tools Used

The following outlines the workstations and all hardware and software used for this investigation.

Workstation

Hostname	Operating System	Build	Physical / Virtual	Built
Windows 11x64_DFIR	Windows 11 Home	2021	Virtual	02/06/2026

Software

Name	Version	Release	Purpose
WinHex/X-ways Forensics	21.6	Oct 19, 2025	A Universal hexadecimal editor, particularly helpful in the realm of computer forensics, data recovery, low-level data processing, and IT security
The Sleuth kit	4.14.0	April 14, 2025	A forensic toolset used to analyze disk images, recover files, and examine file system metadata in digital investigations.
010 Editor	16.0.04	July 31, 2025	A commercial hex editor and text editor developed by SweetScape.

Case Information & Observables

This case involved collecting, processing, and analyzing a Windows-based corporate workstation provided by the Client. The workstation was reported to have had sensitive corporate documents stolen and posted on a competitor's website. The following are my observations and responses to Client questions regarding the disk image, partition layout, filesystem structure, and target file metadata.

Observables

<u>Review the first 446 bytes</u>

1. Does this image have boot code in the first 440 bytes?

Yes, the first 440 bytes contain boot code. This is evident because the bytes are not all zeros and include executable instructions, as well as readable text strings commonly found in MBR code.

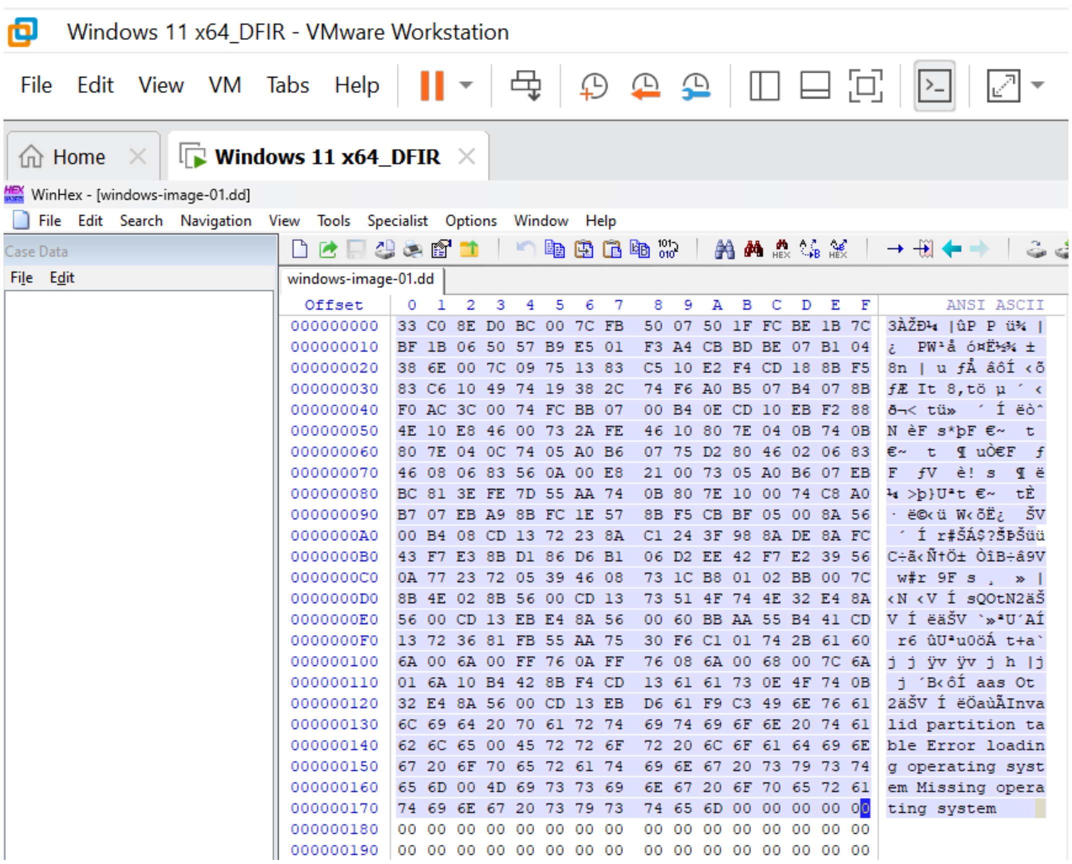


Figure 1: The hexadecimal value of the first 440 bytes of the dd; the plaintext on the side confirms that this is MBR boot code

2. Give your observations including any plaintext in the code.

Within the first 440 bytes, there are two notable plaintext strings: “Invalid partition table”, “Missing operating system”, and “Error loading operating system”. This confirms that the first 440 bytes are of MBR boot code, which is designed to display errors when the system cannot find a valid bootable partition.

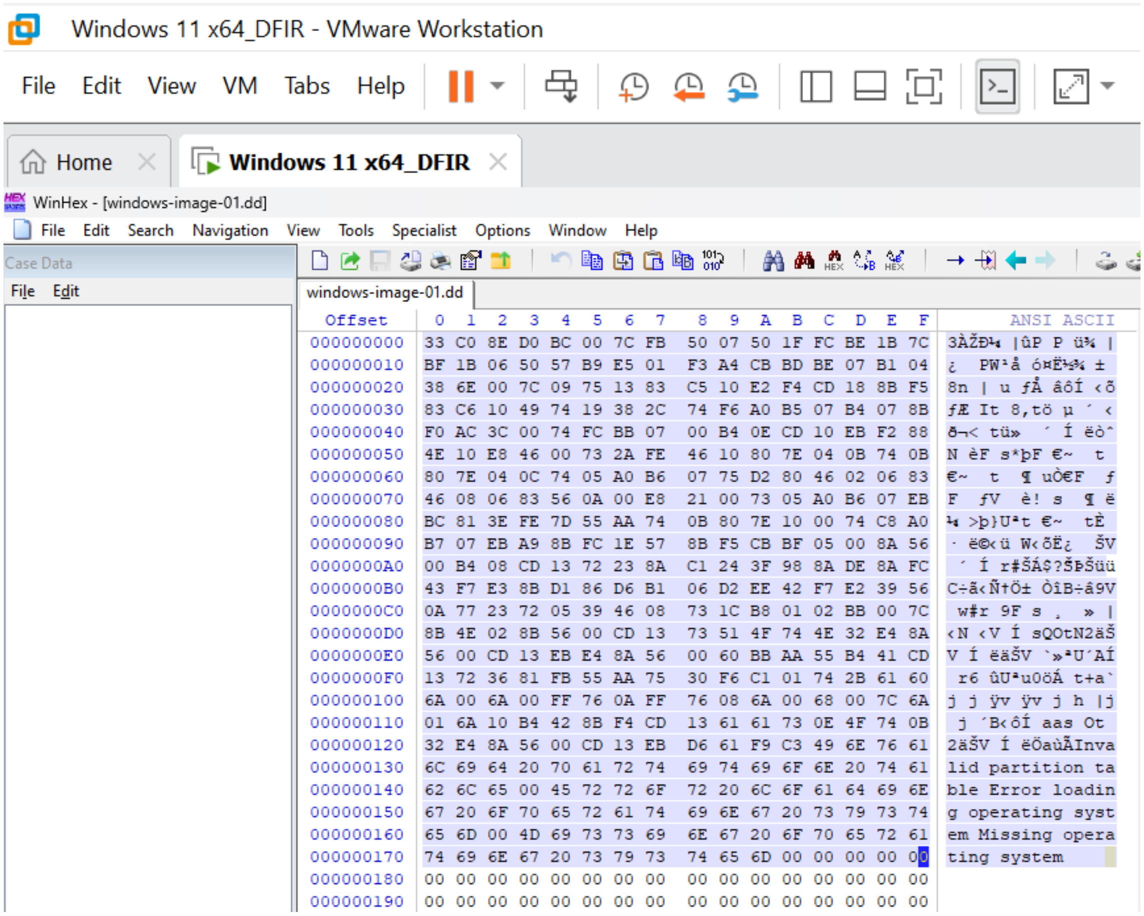


Figure 2: The hexadecimal value of the first 440 bytes of the dd; the plaintext on the side confirms that this is MBR boot code

3. What is the “disk signature” starting at offset 0x01B8? Provide the value as a hexadecimal value like 0x01234567. Give the correct value with endianness in consideration.

The disk signature located at offset 0x01B8 is 0x39BF39BE. Because the data is stored in Little-Endian format, you must read the four bytes at offset 0x01B8 and reverse their order to get the correct hexadecimal representation. This 4-byte identifier is used by the OS to distinguish between physical disks attached to the system.

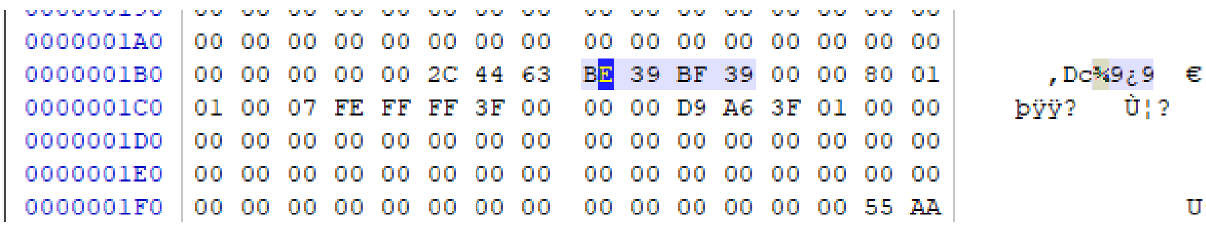


Figure 3: The hexadecimal value of the disk signature starting at offset 0x01B8

Review the partition table beginning at offset 0x01BE

1. How many partition entries are in the table? Include empty ones.

The partition table contains 4 partition entries. This is standard for an MBR, which always allocates space for four entries regardless of whether they are all in use.

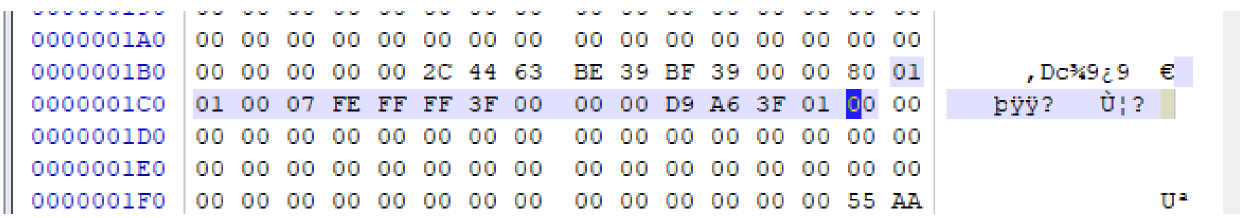


Figure 4: The MBR partition table at offset 0x01BE showing all 4 partition entries

2. Which partition entry is marked bootable?

Partition entry 1 is marked as bootable. This is indicated by the boot value at offset 0x01BE of 0x80 in the first byte of the entry. The partition is active and designated to be used for booting the system, if the value were 0x00 then it would be not bootable.

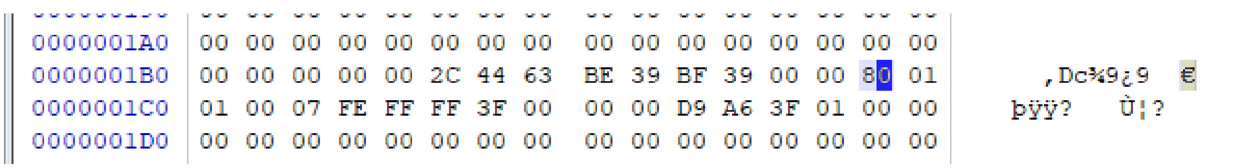


Figure 5: The hexadecimal value at offset 0x01BE showing the value as 80

3. Partition entry 1 - What is the partition entry type? Give the hexadecimal value and the type (e.g. Linux, FAT32)

This partition entry is of type NTFS/exFAT. I found this given the hexadecimal value 0x07 from the value at offset 0x01BE + 4.

```
0000001A0  00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00
0000001B0  00 00 00 00 00 2C 44 63  BE 39 BF 39 00 00 80 01       ,Dc¾9¿9  €
0000001C0  01 00 07 FE FF FF 3F 00  00 00 D9 A6 3F 01 00 00    þÿÿ?    Ù¦?
```

Figure 6: The MBR partition entry 1 showing the value to be 0x07

4. Partition entry 1 - What is the LBA start address? Give the hexadecimal value with endianness in consideration

The LBA start address is located at bytes 8-11 of the partition entry (offset 0x01C6 to 0x01C9). The endianness of the bytes is equivalent to 0x0000003F (which is 63 in decimal).

```
000000170  74 69 6E 67 20 73 79 73  74 65 6D 00 00 00 00 00  ting system
000000180  00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00
000000190  00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00
0000001A0  00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00
0000001B0  00 00 00 00 00 2C 44 63  BE 39 BF 39 00 00 80 01       ,Dc¾9¿9  €
0000001C0  01 00 07 FE FF FF 3F 00  00 00 D9 A6 3F 01 00 00    þÿÿ?    Ù¦?
0000001D0  00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00
0000001E0  00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00
```

Figure 7: The hexadecimal value at offset 0x01BE + 8 which is the LBA start address

5. Calculate the offset of the LBA start address. Sector size is 512 bytes or 0x200 in hexadecimal.

The LBA start for Partition 1 is 0x0000003F which in decimal form is 63. To find the sector size, I multiply 63 *512 = 32,256 (in hexadecimal this is 0x07E00).

6. Partition entry 1 - What is the partition size? Give the hexadecimal value with endianness in consideration

The value of sectors for partition 1 is 0x013FA6D9, which equals 20,948,697 sectors after applying little-endian conversion.

```
000000190  00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00
0000001A0  00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00
0000001B0  00 00 00 00 00 2C 44 63  BE 39 BF 39 00 00 80 01       ,Dc¾9¿9  €
0000001C0  01 00 07 FE FF FF 3F 00  00 00 D9 A6 3F 01 00 00    þÿÿ?    Ù¦?
0000001D0  00 00 00 00 00 00 00 00  00 00 00 00 00 00 00 00
```

Figure 8: The hexadecimal value of Partition 1 at offset 0x01CA-0x01CD

7. Calculate the size of the partition in bytes. Sector size is 512 bytes or 0x200 in hexadecimal Size must be in decimal

The total size of Partition 1 is 10,725,732,864 bytes, calculated as the number of sectors multiplied by 512 bytes per sector.

8. What is the difference, in bytes, between the actual size of the evidence file versus the value above?

The difference between the total disk size and the calculated partition size is 11,685,376 bytes, calculated by the number of bytes in the dd image minus the size of the partition in bytes. This difference represents space occupied by the MBR, unused sectors, and any padding within the disk image.

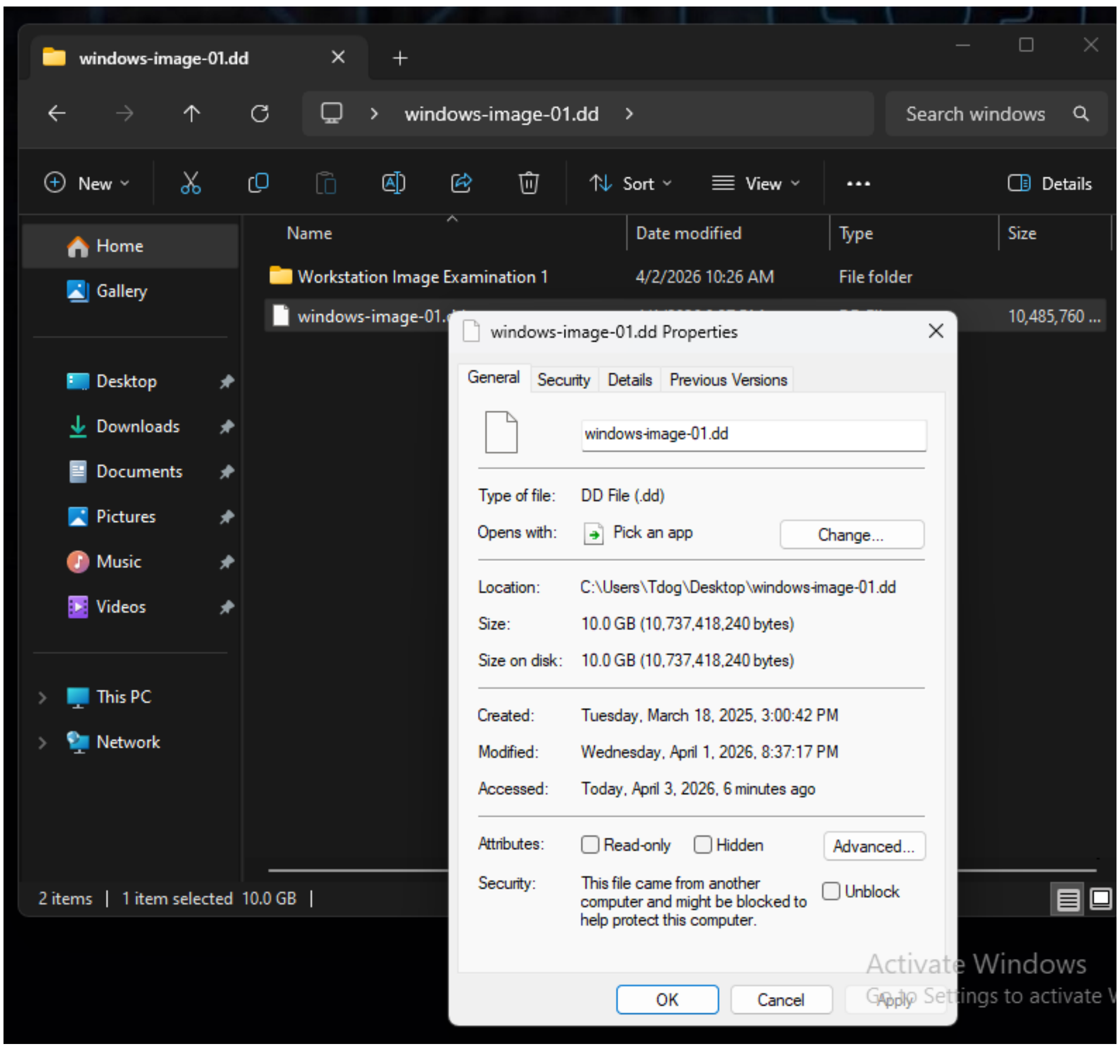


Figure 9: Screenshot of the size of the DD file

Navigate to Partition Entry One LBA Start Address Offset

1. What are the first 8 bytes at this offset? DO NOT consider endianness in this value

The first 8 bytes at the partition's starting offet (0x01BE) are: 80 01 01 00 07 FE FF FF.

2. Does this value support your partition type answer above?

Yes, because the byte at offset 0x01C2 is 07 which is the standardized Hex code for the NTFS/exFAT file system. This is a commonly used file system in Windows environments.

3. Give your observations including any plaintext in the code.

The hex dump contains standard MBR partition table fields, including a boot indicator, LBA start/end address, and the partition type. To the right of the hex code, you can see portions of an operating system string: "Missing operating system". This is a common error message found in MBR bootloader code areas.

4. Based on your observations, does this disk image use an MBR or GPT partitioning scheme? Explain why.

This disk image uses an MBR partitioning scheme. The data is structured as 16-byte partition entries starting at 0x01BE. This is the exact layout of the legacy MBR partition table. A GPT partition entry is 128 bytes in size.

Examination 2 Questions

1. Using 'mmls' , what is the starting sector of the NTFS/exFAT partition?

The starting sector of the NTFS/exFAT partition is 63, as identified using the 'mmls' command. This value matches the LBA start addressed determined during the previous WinHex analysis, where the starting sector was calculated as 0x0000003F, which is equivalent to 63 in decimal. This starting sector tells us where our partition should start for when we do further analysis.

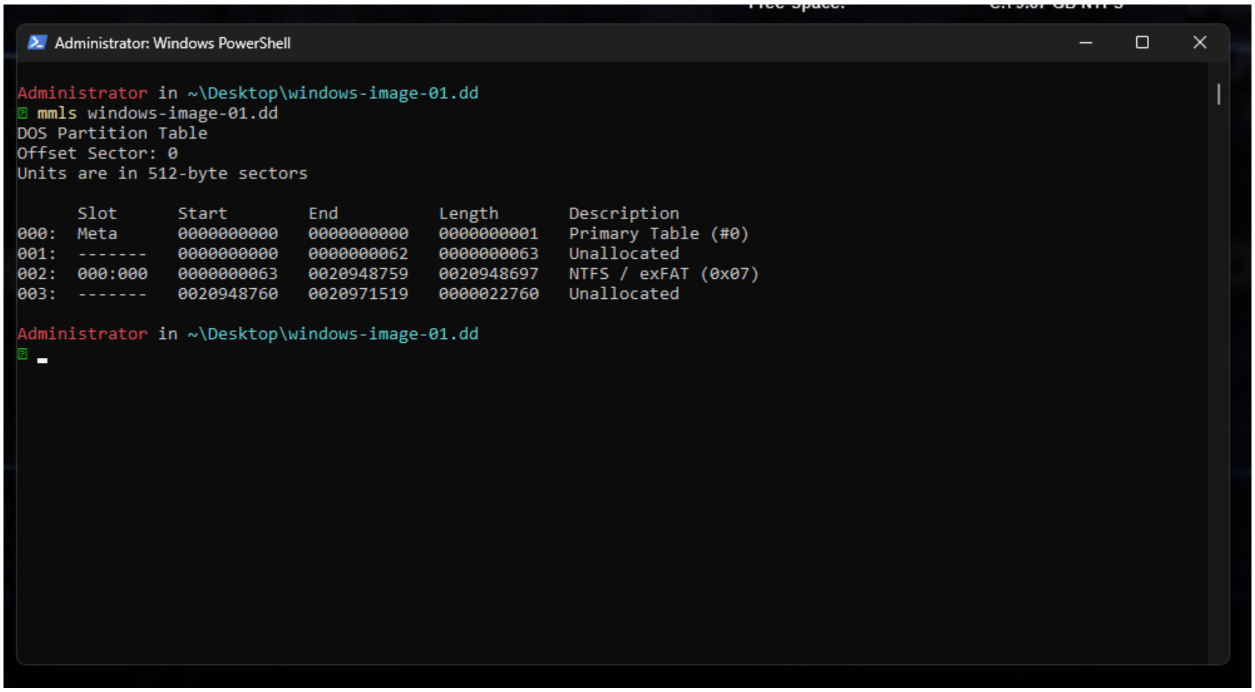


Figure 10: The Sleuth Kit 'mmls' command to find starting sector

2. Using ‘fsstat’ , what is the volume serial number?

Using ‘fsstat’ I was able to find file system information and confirm that the file system was NTFS, which we already knew from the previous examination. The volume serial number was 7E745008744FC21F and this version was “Windows XP”.

Figure 11: The Sleuth Kit 'fsstat' command to find volume serial number

3. What is the sector size?

Using the ‘fsstat’ command, I was able to find the sector size to be 512 bytes. A sector is one tiny block on a disk, it is the smallest addressable unit on the physical disk used by a filesystem.

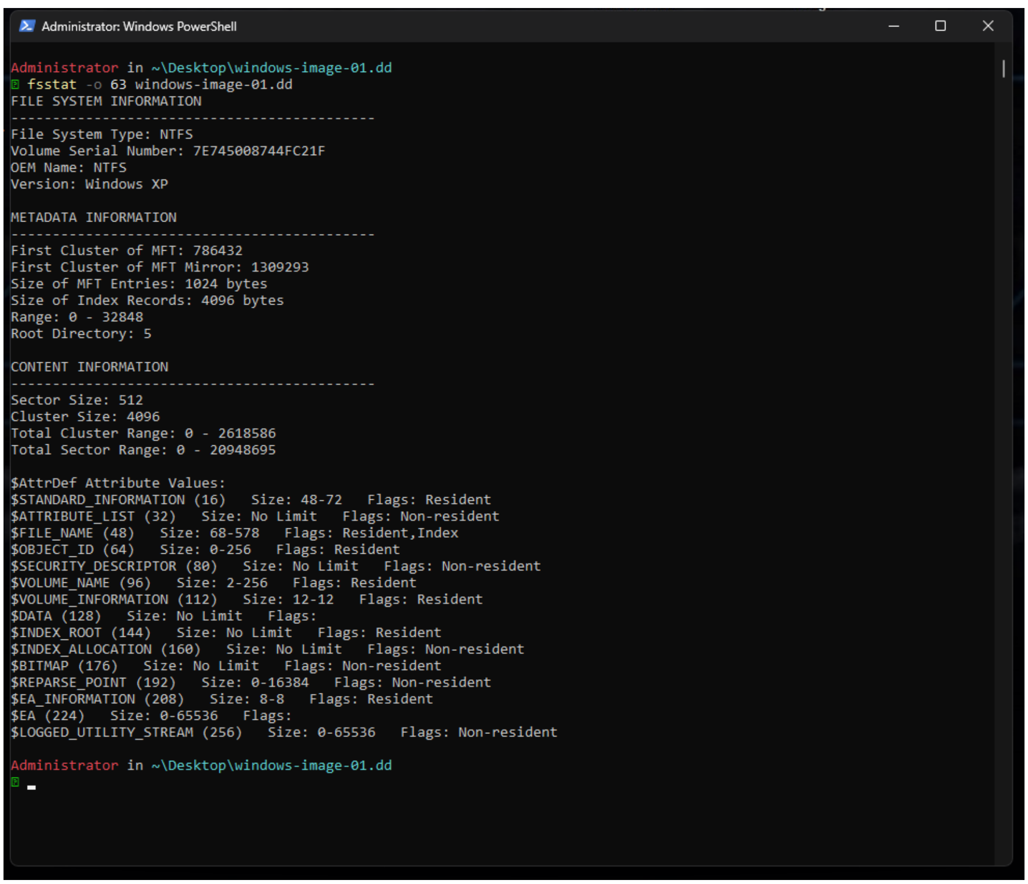


Figure 12: The Sleuth Kit 'fsstat' command to find sector size and cluster size

4. What is the cluster size?

Using the ‘fsstat’ command, I was able to find the cluster size to be 4096 bytes. This matches the sector size of 512 bytes because 8 sectors for a cluster (512 * 8 = 4096), which is consistent with NTFS allocation behavior.

5. How many sectors fit into one cluster?

There are 8 sectors that belong to 1 cluster. A cluster consists of multiple sectors, allowing the filesystem to manage storage more efficiently. Therefore, if the sector size is 512 bytes, the cluster size should be 512 * 8 = 4096 bytes.

6. What is the first cluster of the MFT?

Using the ‘fsstat’ command, I was able to find the MFT first cluster to be 786432. The Master File Table (MFT) is a critical structure in NTFS that stores metadata for every file and directory on the system.

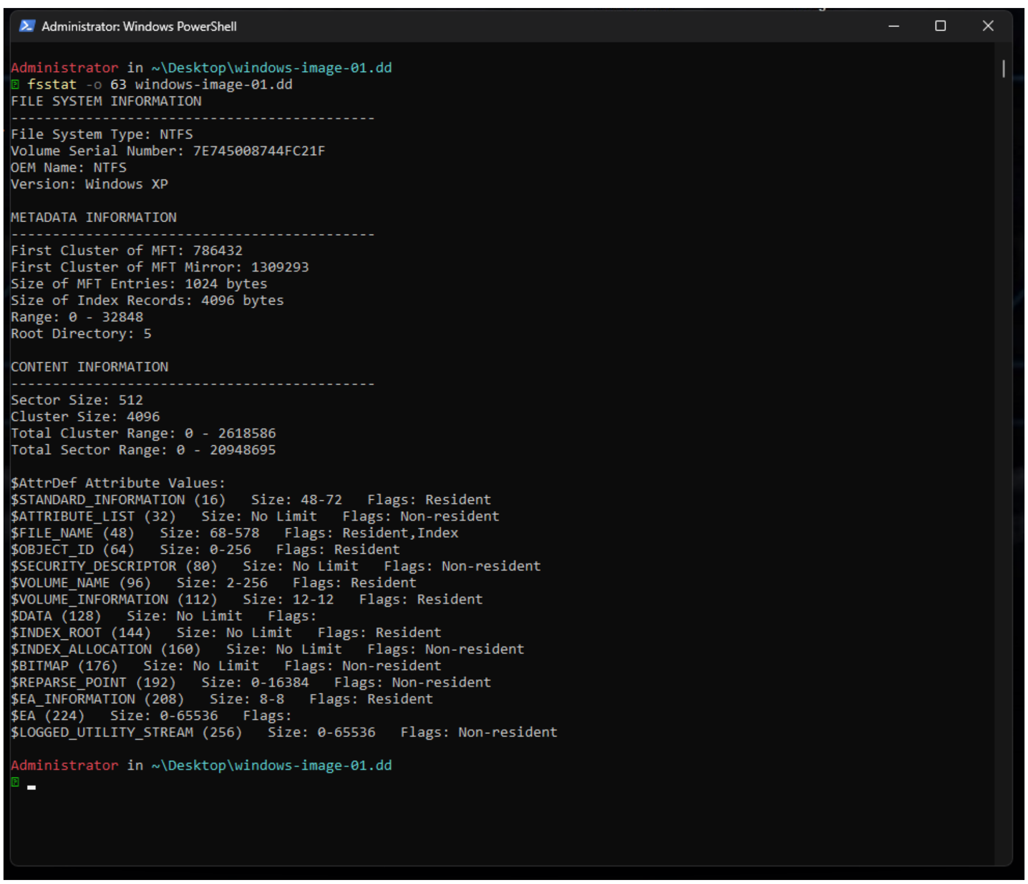
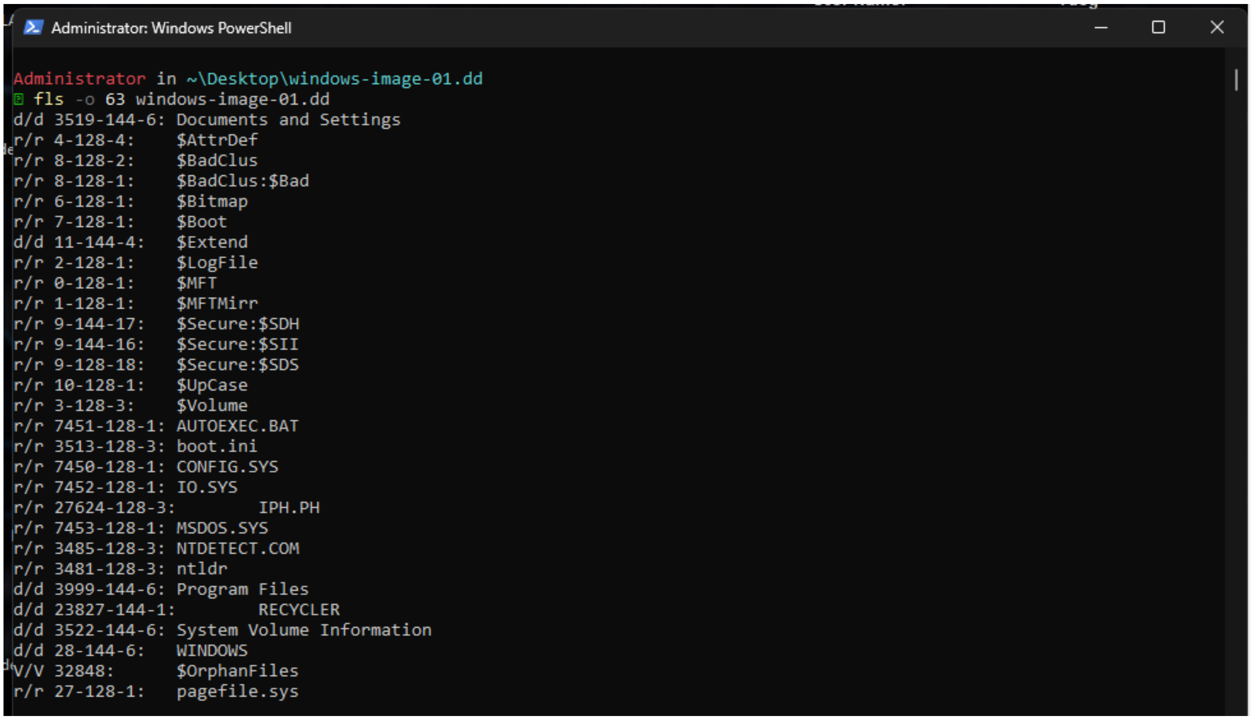


Figure 13: The Sleuth Kit 'fsstat' command to find metadata information

7. **Using ‘fls’, what is the metadata reference (inode) for the folder “Documents and Settings”?**

The metadata reference for the folder “Documents and Settings” is inode 3519 and is a directory, as identified using the ‘fls’ command.



```
Administrator: Windows PowerShell

Administrator in ~\Desktop\windows-image-01.dd
fls -o 63 windows-image-01.dd
d/d 3519-144-6: Documents and Settings
r/r 4-128-4:    $AttrDef
r/r 8-128-2:    $BadClus
r/r 8-128-1:    $BadClus:$Bad
r/r 6-128-1:    $Bitmap
r/r 7-128-1:    $Boot
d/d 11-144-4:   $Extend
r/r 2-128-1:    $LogFile
r/r 0-128-1:    $MFT
r/r 1-128-1:    $MFTMirr
r/r 9-144-17:   $Secure:$SDH
r/r 9-144-16:   $Secure:$SII
r/r 9-128-18:   $Secure:$SDS
r/r 10-128-1:   $UpCase
r/r 3-128-3:    $Volume
r/r 7451-128-1: AUTOEXEC.BAT
r/r 3513-128-3: boot.ini
r/r 7450-128-1: CONFIG.SYS
r/r 7452-128-1: IO.SYS
r/r 27624-128-3:        IPH.PH
r/r 7453-128-1: MSDOS.SYS
r/r 3485-128-3: NTDETECT.COM
r/r 3481-128-3: ntldr
d/d 3999-144-6: Program Files
d/d 23827-144-1:        RECYCLER
d/d 3522-144-6: System Volume Information
d/d 28-144-6:   WINDOWS
V/V 32848:      $OrphanFiles
r/r 27-128-1:   pagefile.sys
```

Figure 14: The Sleuth Kit 'fls' command to find the “Documents and Settings” folder

8. **Using this metadata reference and ‘fls’ , there are three directories representing interactive users (users that have logged into the system). What are the directory (user) names? Do not include “All Users”, “Default User”, “Local Service”, or “Network Service”.**

Using the metadata reference for the “Documents and Settings” directory (inode 3519), the ‘fls’ command was used to enumerate its contents. The interactive user directories identified were **Administrator (inode 10222), Devon (inode 17437),** and **Jean (inode 16144)**. All of the users are identified as directories.

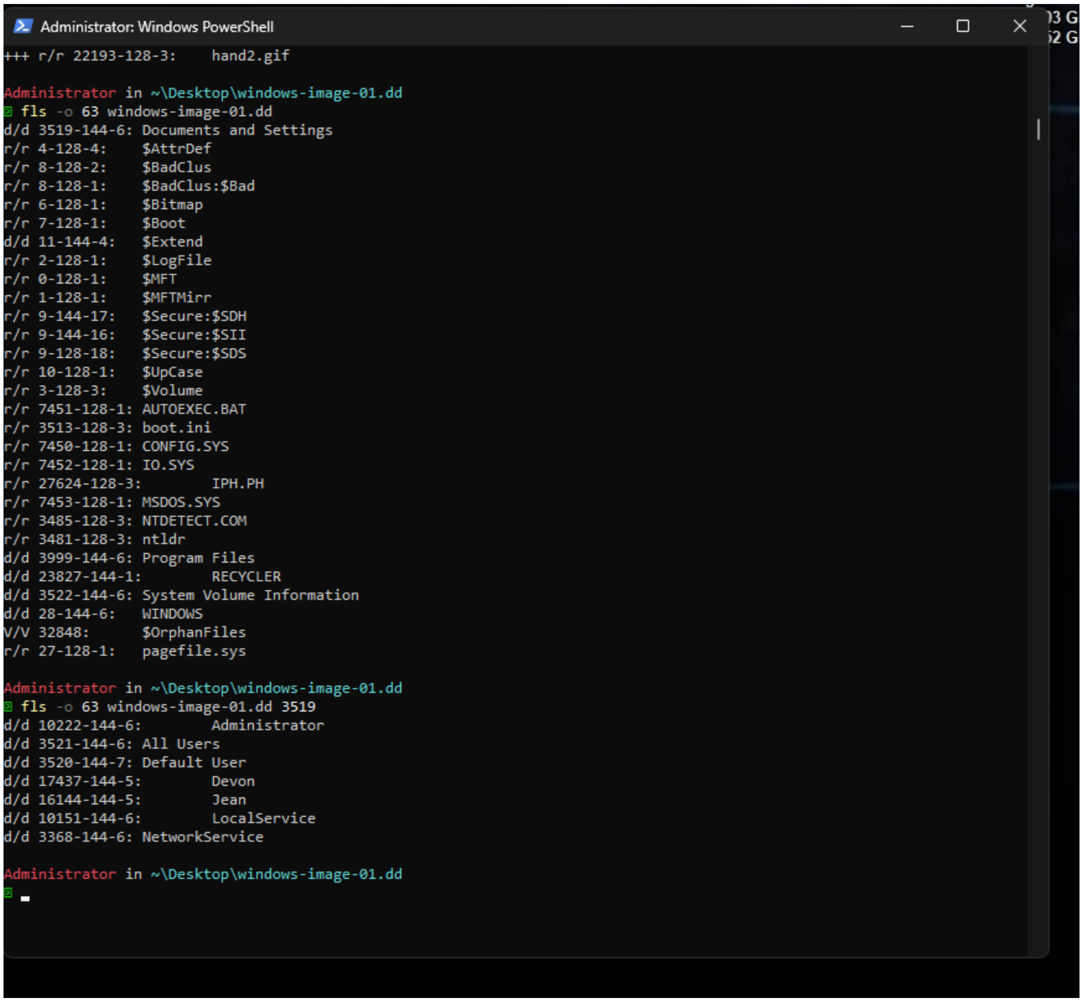


Figure 15: The Sleuth Kit 'fls' command from the "Documents and Settings" directory to find users

9. **Using 'istat' and the metadata reference for the folder "Devon", review the $STANDARD_INFORMATION block. What is the create date timestamp?**

Using the 'istat' command on the metadata reference for the folder "Devon", the created timestamp from the "$STANDARD_INFORMATION" attribute is 2008-07-11 20:02:48.006383500 (Pacific Daylight Time).

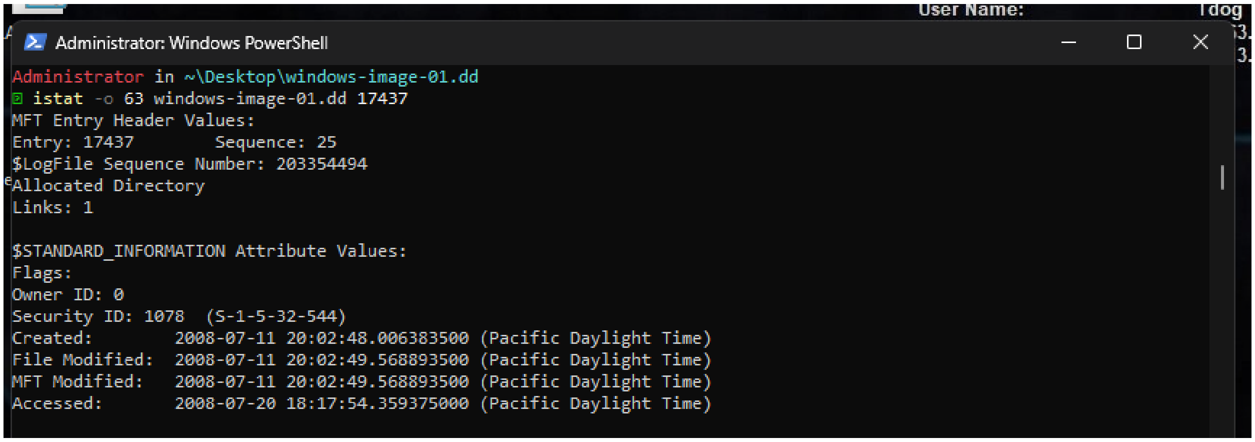


Figure 16: The Sleuth Kit 'istat' command used to obtain information about the user 'Devon'

10. Do the same for the folder "Jean". What is the create date timestamp?

Using the 'istat' command on the metadata reference for the folder "Jean", the created timestamp from the "$STANDARD_INFORMATION" attribute is 2008-07-05 23:11:22.238875000 (Pacific Daylight Time).

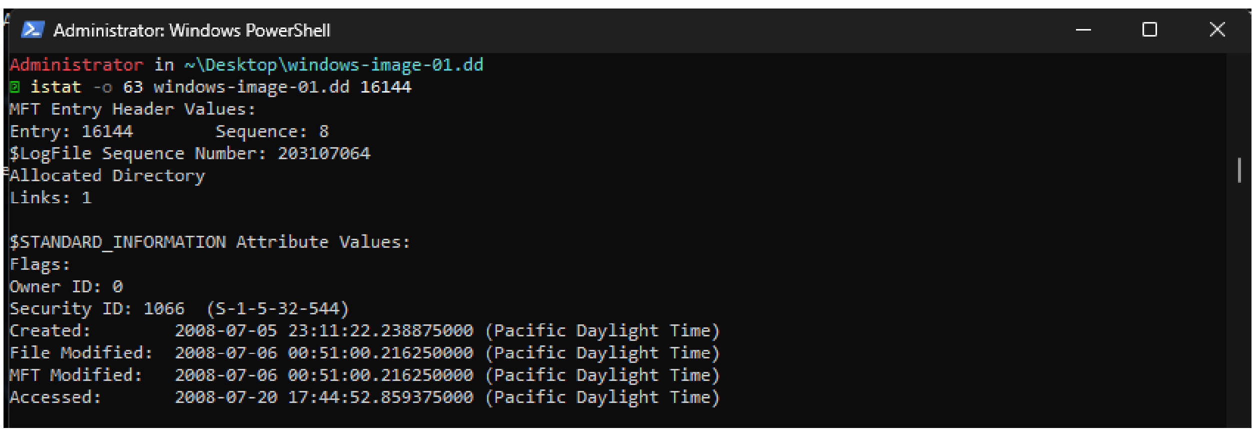


Figure 17: The Sleuth Kit 'istat' command used to obtain information about the user 'Jean'

11. Based on this information, what user profile was created first?

Based on the created timestamps from the users, "Jean" was created 6 days before "Devon".

12. Using ‘fls’ , review the contents of Jean’s “Desktop” folder. What are the file names of the two files?

Using the ‘fls’ command on the metadata reference for the “Jean” user directory, the contents of the Desktop folder were examined. The two files present in the Desktop directory are ‘AIM Tunes.url’ and ‘m57biz.xls’.

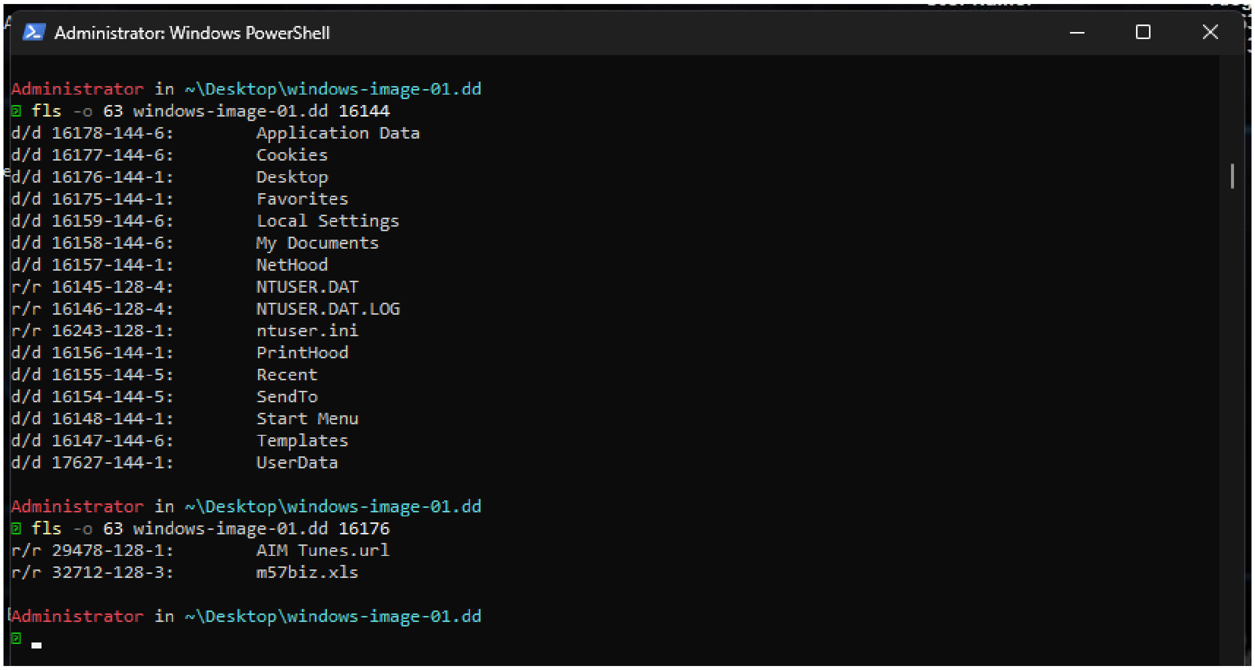
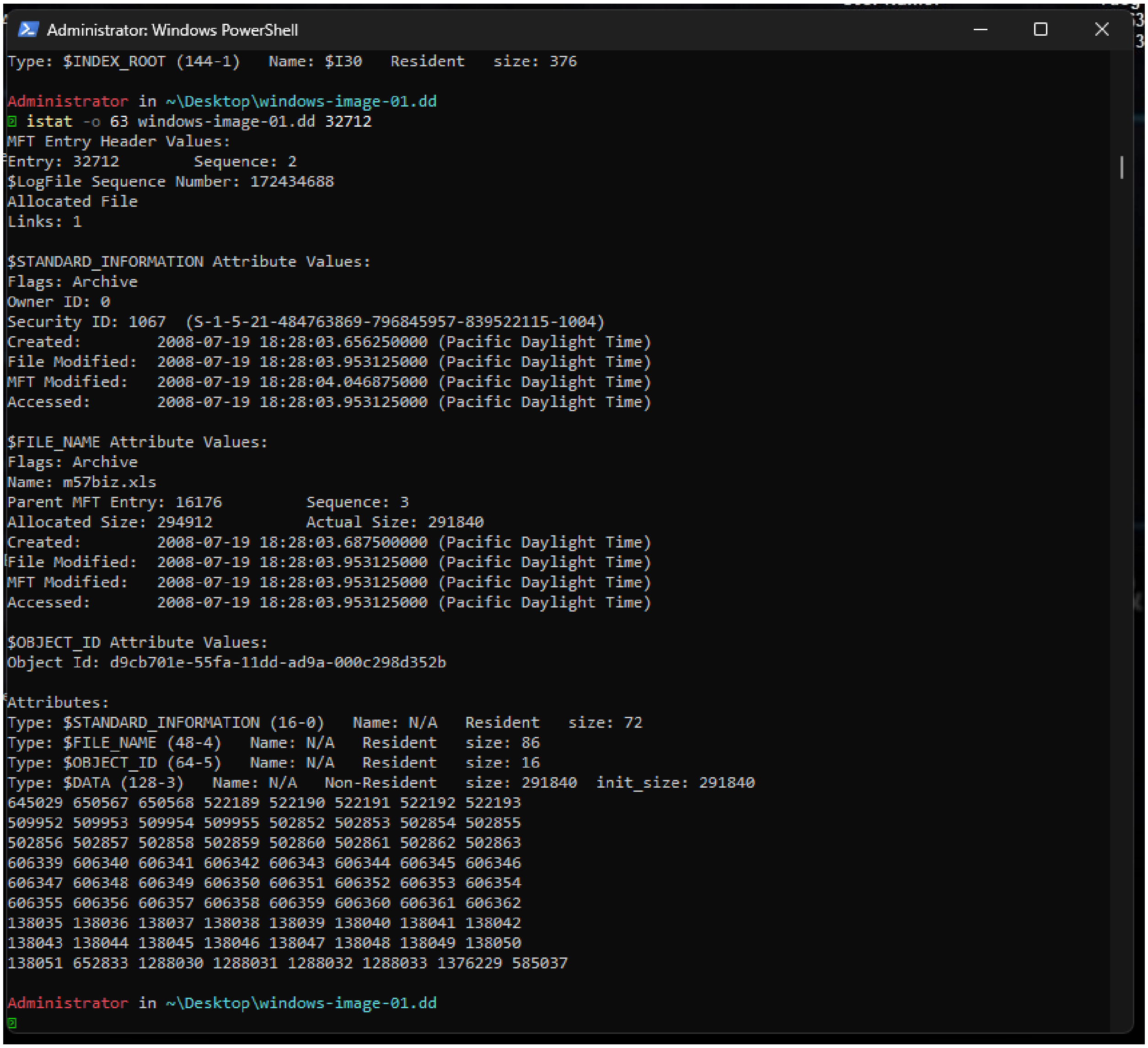


Figure 18: The Sleuth Kit 'fls' command used to view the contents of the Desktop directory

13. Using ‘istat’ , what is the created date/timestamp of the file WITHOUT the “.url” extension?

Using the ‘istat’ command on the file without the “.url" extension, the created timestamp from the $STANDARD_INFORMATION attribute is 2008-07-19 18:28:03.656250000.



```
Administrator: Windows PowerShell
Type: $INDEX_ROOT (144-1)   Name: $I30   Resident   size: 376

Administrator in ~\Desktop\windows-image-01.dd
 istat -o 63 windows-image-01.dd 32712
MFT Entry Header Values:
Entry: 32712        Sequence: 2
$LogFile Sequence Number: 172434688
Allocated File
Links: 1

$STANDARD_INFORMATION Attribute Values:
Flags: Archive
Owner ID: 0
Security ID: 1067  (S-1-5-21-484763869-796845957-839522115-1004)
Created:        2008-07-19 18:28:03.656250000 (Pacific Daylight Time)
File Modified:  2008-07-19 18:28:03.953125000 (Pacific Daylight Time)
MFT Modified:   2008-07-19 18:28:04.046875000 (Pacific Daylight Time)
Accessed:       2008-07-19 18:28:03.953125000 (Pacific Daylight Time)

$FILE_NAME Attribute Values:
Flags: Archive
Name: m57biz.xls
Parent MFT Entry: 16176         Sequence: 3
Allocated Size: 294912          Actual Size: 291840
Created:        2008-07-19 18:28:03.687500000 (Pacific Daylight Time)
File Modified:  2008-07-19 18:28:03.953125000 (Pacific Daylight Time)
MFT Modified:   2008-07-19 18:28:03.953125000 (Pacific Daylight Time)
Accessed:       2008-07-19 18:28:03.953125000 (Pacific Daylight Time)

$OBJECT_ID Attribute Values:
Object Id: d9cb701e-55fa-11dd-ad9a-000c298d352b

Attributes:
Type: $STANDARD_INFORMATION (16-0)   Name: N/A   Resident   size: 72
Type: $FILE_NAME (48-4)   Name: N/A   Resident   size: 86
Type: $OBJECT_ID (64-5)   Name: N/A   Resident   size: 16
Type: $DATA (128-3)   Name: N/A   Non-Resident   size: 291840  init_size: 291840
645029 650567 650568 522189 522190 522191 522192 522193
509952 509953 509954 509955 502852 502853 502854 502855
502856 502857 502858 502859 502860 502861 502862 502863
606339 606340 606341 606342 606343 606344 606345 606346
606347 606348 606349 606350 606351 606352 606353 606354
606355 606356 606357 606358 606359 606360 606361 606362
138035 138036 138037 138038 138039 138040 138041 138042
138043 138044 138045 138046 138047 138048 138049 138050
138051 652833 1288030 1288031 1288032 1288033 1376229 585037

Administrator in ~\Desktop\windows-image-01.dd
```

Figure 19: The Sleuth Kit 'istat' command used to obtain information about the m57biz.xls file

14. Are there any files in Devon's "Desktop" folder? If so, what are the file names?

Using the 'fls' command to examine the contents of Devon's "Desktop" directory (inode 23955), no files or directories were identified. Therefore, the Desktop folder for the user Devon does not contain any files.

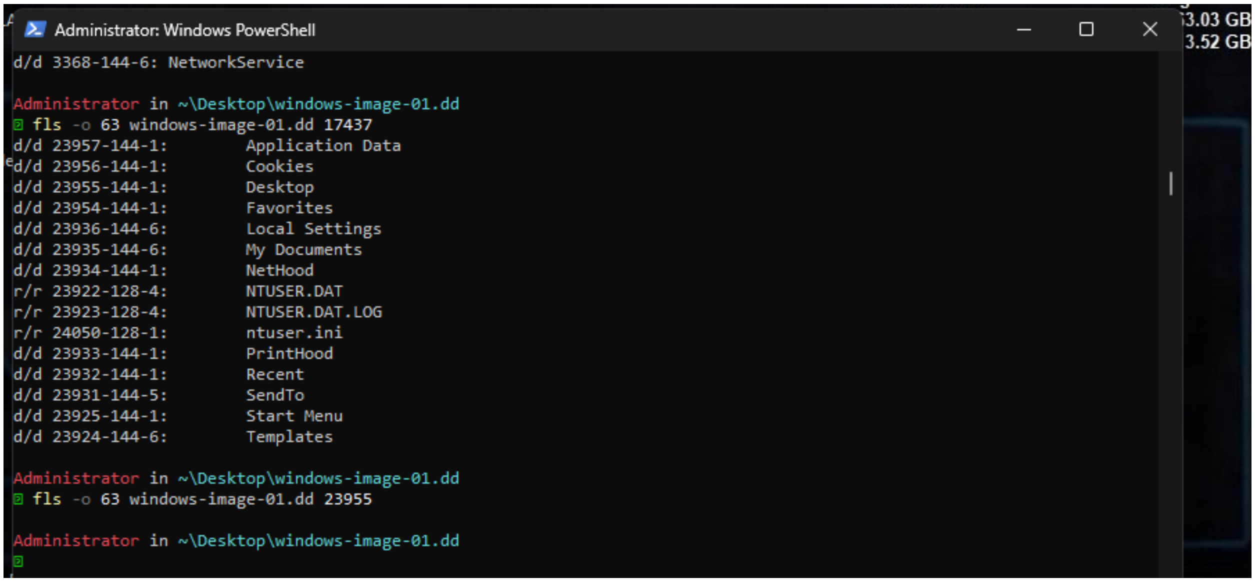


Figure 20: The Sleuth Kit 'fls' command used to find the contents of the Desktop directory for "Devon"

15. Use 'icat' to recover the file from question 13 to disk. Run the 'file' command on the recovered file. According to the file output, who was the "author" and "last saved" by entries?

Using the 'icat' command, the file identified in Question 13 was successfully recovered from the disk image and written to disk. The recovered file was analyzed using string extraction to identify embedded metadata. Based on the output, the 'author' of the file is **Alison Smith**, and the 'last saved by' entry is **Jean**.

```
Alison Smith
lis
lis
Jean User
Microsoft Excel
 `%Z%
" WMFC
 EMF
j%v
0`v
j%#
j%#
0dv
<%Q%A
<%Q%A
<%Q%A
<%Q%A
<%Q%A
<%Q%A
<%Q%A
<%Q%A
<%Q%A
<%Q%A
<%Q%A
<%Q%A
<%Q%A
WMFC
<%Q%A
<%Q%A
<%Q%A
<%Q%A
<%Q%A
<%Q%A
<%Q%A
<%Q%A
<%Q%A
<%Q%A
<%Q%A
<%Q%A
Verdana
"System
M57.biz company
Name
Position
Salary
Alison
Smithn
Presidento
$140,000
Jean
Jones0
CFO
$120,000
R%P%R%
R%P%R%
M57.BIZ
Sheet1
Worksheets
_PID_LINKBASE
j%K

Administrator in ~\Desktop\windows-image-01.dd took 4s
```

Figure 21: Using the 'strings' command to

Examination 3 Questions

1. **Does the target file m57biz.xls exist on this system? Does it have the same content as the exfiled content?**

   The file *m57biz.xls* was identified on the disk image using the 'fls' command with recursive listing, confirming its existence within the file system. The file was then recovered using the 'icat' command and analyzed. A comparison between the recovered file and the data observed on the competitor's website indicates that the contents are consistent, and the image is the same as the exfiled content.

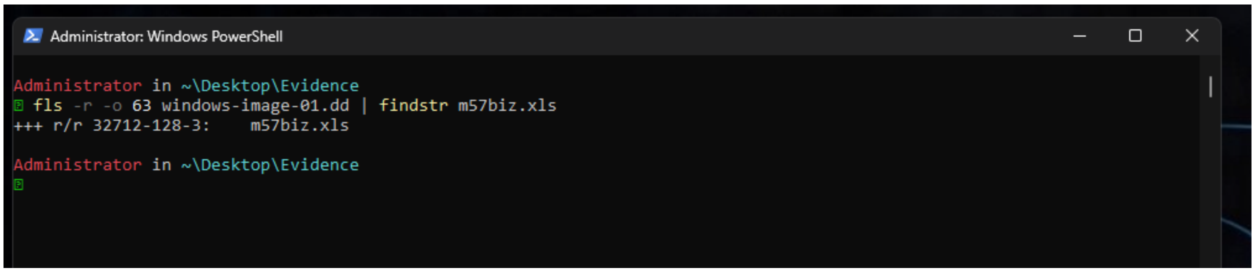


Figure 22: TSK 'fls' command used to find 'm57biz.xls'

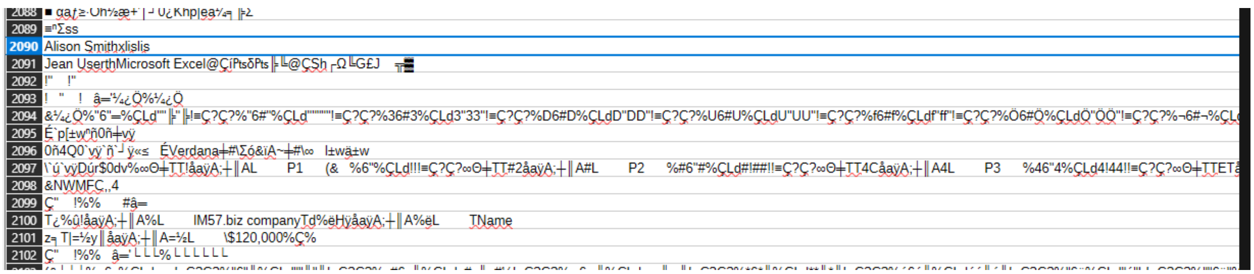


Figure 23: Concatenating 'm57biz.xls' to a file and comparing the contents to exfiled content

2. **Using ffind, what directory does the target file reside in?**

   The file '*m57biz.xls'* was identified with 'fls' command and 'ffind' command found the directory that the file resided in. The output shows the exact directory.

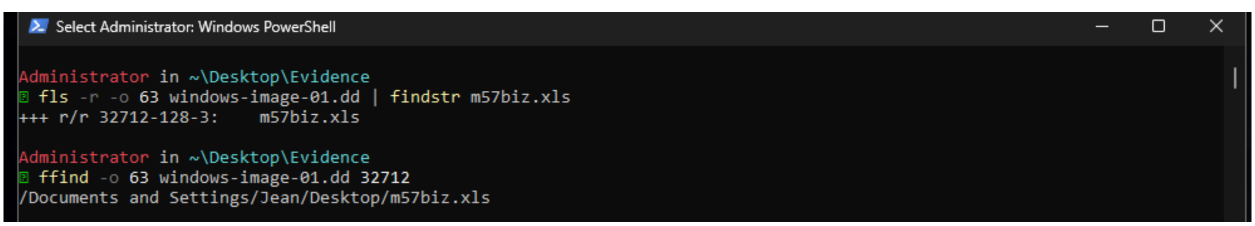


Figure 24: TSK using 'ffind' command to show the directory of the 'm57biz.xls' file

3. **What is the metadata reference for the target file?**

The metadata reference for the target file 'm57biz.xls' is 32712, as identified using the 'istat' command. This value corresponds to the MFT entry associated with the file within the file system.

```
Administrator in ~\Desktop\Evidence
 istat -o 63 windows-image-01.dd 32712
MFT Entry Header Values:
Entry: 32712        Sequence: 2
$LogFile Sequence Number: 172434688
Allocated File
Links: 1

$STANDARD_INFORMATION Attribute Values:
Flags: Archive
Owner ID: 0
Security ID: 1067  (S-1-5-21-484763869-796845957-839522115-1004)
Created:        2008-07-19 18:28:03.656250000 (Pacific Daylight Time)
File Modified:  2008-07-19 18:28:03.953125000 (Pacific Daylight Time)
MFT Modified:   2008-07-19 18:28:04.046875000 (Pacific Daylight Time)
Accessed:       2008-07-19 18:28:03.953125000 (Pacific Daylight Time)

$FILE_NAME Attribute Values:
Flags: Archive
Name: m57biz.xls
Parent MFT Entry: 16176         Sequence: 3
Allocated Size: 294912          Actual Size: 291840
Created:        2008-07-19 18:28:03.687500000 (Pacific Daylight Time)
File Modified:  2008-07-19 18:28:03.953125000 (Pacific Daylight Time)
MFT Modified:   2008-07-19 18:28:03.953125000 (Pacific Daylight Time)
Accessed:       2008-07-19 18:28:03.953125000 (Pacific Daylight Time)

$OBJECT_ID Attribute Values:
Object Id: d9cb701e-55fa-11dd-ad9a-000c298d352b

Attributes:
Type: $STANDARD_INFORMATION (16-0)   Name: N/A   Resident   size: 72
Type: $FILE_NAME (48-4)   Name: N/A   Resident   size: 86
Type: $OBJECT_ID (64-5)   Name: N/A   Resident   size: 16
Type: $DATA (128-3)   Name: N/A   Non-Resident   size: 291840  init_size: 291840
645029 650567 650568 522189 522190 522191 522192 522193
509952 509953 509954 509955 502852 502853 502854 502855
502856 502857 502858 502859 502860 502861 502862 502863
606339 606340 606341 606342 606343 606344 606345 606346
606347 606348 606349 606350 606351 606352 606353 606354
606355 606356 606357 606358 606359 606360 606361 606362
138035 138036 138037 138038 138039 138040 138041 138042
138043 138044 138045 138046 138047 138048 138049 138050
138051 652833 1288030 1288031 1288032 1288033 1376229 585037

Administrator in ~\Desktop\Evidence
```

Figure 25: TSK 'istat' command shows the metadata for 'm57biz.xls'

4. **What is the MFT entry for this file. (Hint: its the first number in the fls metadata reference)**

The MFT entry for this file is 32712 which is consistent with the metadata reference. The first number in the 'fls' metadata reference is the same as the MFT entry header value from the 'istat' command.

```
Administrator: Windows PowerShell

Administrator in ~\Desktop\Evidence
❯ fls -r -o 63 windows-image-01.dd | findstr m57biz.xls
+++ r/r 32712-128-3:    m57biz.xls

Administrator in ~\Desktop\Evidence
❯ icat -o 63 windows-image-01.dd 32712 > m57biz.xls

Administrator in ~\Desktop\Evidence
❯ istat -o 63 windows-image-01.dd 32712
MFT Entry Header Values:
Entry: 32712        Sequence: 2
$LogFile Sequence Number: 172434688
Allocated File
Links: 1

$STANDARD_INFORMATION Attribute Values:
```

Figure 26: TSK 'fls' and 'istat' command show the MFT entry for ‘m57biz.xls’

5. **Based on this information, what is the offset in the MFT for this entry. (Hint: this number should be in the 30 million)**

   The MFT entry for ‘m57biz.xls’ is 32712. Since each MFT entry is 1024 bytes, the offset fotr this entry within the MFT is calculated as 32712 * 1024 = 33,497,088 bytes. This places the file’s MFT entry at the offset 33,497,088.

6. **Review the MFT for the target file, what is the length of $STANDARD_INFO in bytes?**

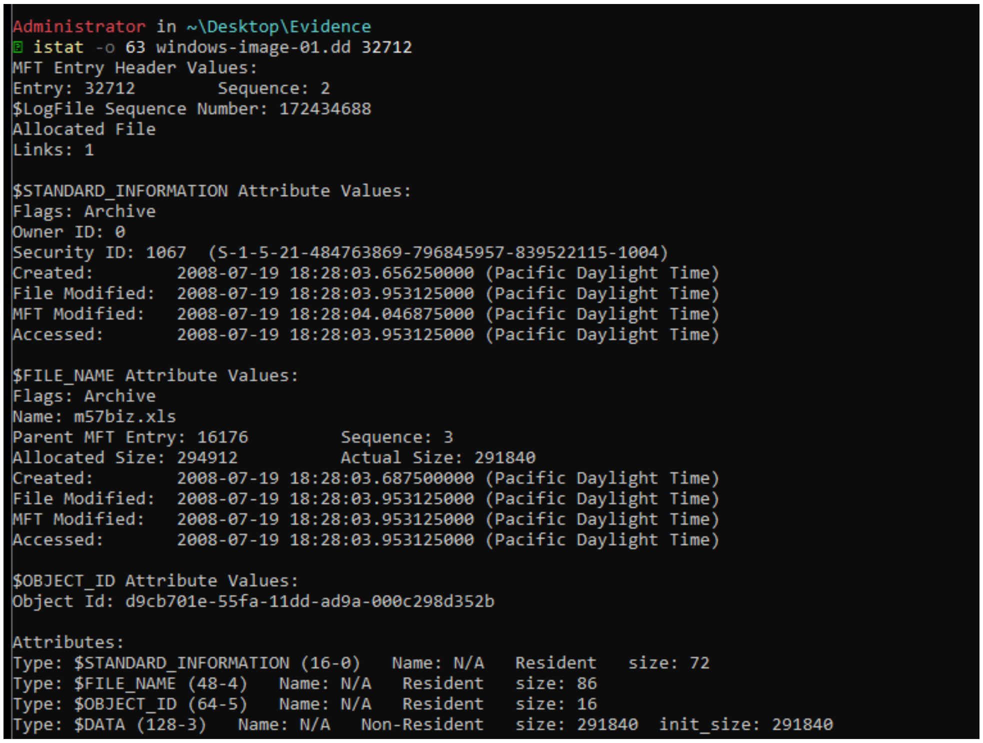
   The length of the $STANDARD_INFORMATION attribute for the file ‘m57biz.xls’ is 72 bytes, as identified using the ‘istat’ command.

```
Administrator in ~\Desktop\Evidence
❯ istat -o 63 windows-image-01.dd 32712
MFT Entry Header Values:
Entry: 32712        Sequence: 2
$LogFile Sequence Number: 172434688
Allocated File
Links: 1

$STANDARD_INFORMATION Attribute Values:
Flags: Archive
Owner ID: 0
Security ID: 1067  (S-1-5-21-484763869-796845957-839522115-1004)
Created:        2008-07-19 18:28:03.656250000 (Pacific Daylight Time)
File Modified:  2008-07-19 18:28:03.953125000 (Pacific Daylight Time)
MFT Modified:   2008-07-19 18:28:04.046875000 (Pacific Daylight Time)
Accessed:       2008-07-19 18:28:03.953125000 (Pacific Daylight Time)

$FILE_NAME Attribute Values:
Flags: Archive
Name: m57biz.xls
Parent MFT Entry: 16176        Sequence: 3
Allocated Size: 294912         Actual Size: 291840
Created:        2008-07-19 18:28:03.687500000 (Pacific Daylight Time)
File Modified:  2008-07-19 18:28:03.953125000 (Pacific Daylight Time)
MFT Modified:   2008-07-19 18:28:03.953125000 (Pacific Daylight Time)
Accessed:       2008-07-19 18:28:03.953125000 (Pacific Daylight Time)

$OBJECT_ID Attribute Values:
Object Id: d9cb701e-55fa-11dd-ad9a-000c298d352b

Attributes:
Type: $STANDARD_INFORMATION (16-0)   Name: N/A   Resident   size: 72
Type: $FILE_NAME (48-4)   Name: N/A   Resident   size: 86
Type: $OBJECT_ID (64-5)   Name: N/A   Resident   size: 16
Type: $DATA (128-3)   Name: N/A   Non-Resident   size: 291840  init_size: 291840
```

Figure 27: TSK using 'istat' command to show metadata of $STANDARD_INFORMATION

7. **What is the length of $FILE_NAME in bytes?**

The length of the $FILE_NAME attribute for the file 'm57biz.xls' is 86 bytes, as identified using the 'istat' command.

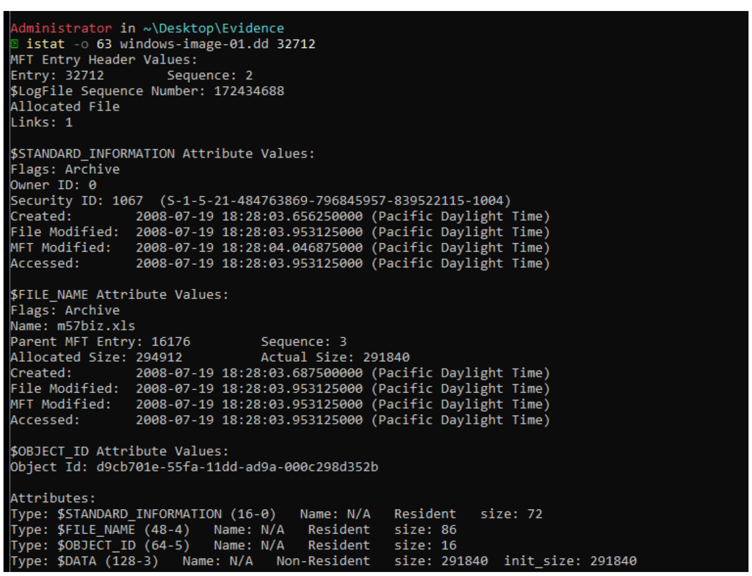


```
Administrator in ~\Desktop\Evidence
istat -o 63 windows-image-01.dd 32712
MFT Entry Header Values:
Entry: 32712        Sequence: 2
$LogFile Sequence Number: 172434688
Allocated File
Links: 1

$STANDARD_INFORMATION Attribute Values:
Flags: Archive
Owner ID: 0
Security ID: 1067  (S-1-5-21-484763869-796845957-839522115-1004)
Created:        2008-07-19 18:28:03.656250000 (Pacific Daylight Time)
File Modified:  2008-07-19 18:28:03.953125000 (Pacific Daylight Time)
MFT Modified:   2008-07-19 18:28:04.046875000 (Pacific Daylight Time)
Accessed:       2008-07-19 18:28:03.953125000 (Pacific Daylight Time)

$FILE_NAME Attribute Values:
Flags: Archive
Name: m57biz.xls
Parent MFT Entry: 16176         Sequence: 3
Allocated Size: 294912          Actual Size: 291840
Created:        2008-07-19 18:28:03.687500000 (Pacific Daylight Time)
File Modified:  2008-07-19 18:28:03.953125000 (Pacific Daylight Time)
MFT Modified:   2008-07-19 18:28:03.953125000 (Pacific Daylight Time)
Accessed:       2008-07-19 18:28:03.953125000 (Pacific Daylight Time)

$OBJECT_ID Attribute Values:
Object Id: d9cb701e-55fa-11dd-ad9a-000c298d352b

Attributes:
Type: $STANDARD_INFORMATION (16-0)   Name: N/A   Resident   size: 72
Type: $FILE_NAME (48-4)   Name: N/A   Resident   size: 86
Type: $OBJECT_ID (64-5)   Name: N/A   Resident   size: 16
Type: $DATA (128-3)   Name: N/A   Non-Resident   size: 291840  init_size: 291840
```

Figure 28: TSK using 'istat' command to show metadata of $FILE_NAME

8. **Is there another attribute after $FILE_NAME and before $DATA? What is it?**

There is another attribute after $FILE_NAME and it is $OBJECT_ID, as identified using the 'istat' command.



```
Administrator in ~\Desktop\Evidence
istat -o 63 windows-image-01.dd 32712
MFT Entry Header Values:
Entry: 32712        Sequence: 2
$LogFile Sequence Number: 172434688
Allocated File
Links: 1

$STANDARD_INFORMATION Attribute Values:
Flags: Archive
Owner ID: 0
Security ID: 1067  (S-1-5-21-484763869-796845957-839522115-1004)
Created:        2008-07-19 18:28:03.656250000 (Pacific Daylight Time)
File Modified:  2008-07-19 18:28:03.953125000 (Pacific Daylight Time)
MFT Modified:   2008-07-19 18:28:04.046875000 (Pacific Daylight Time)
Accessed:       2008-07-19 18:28:03.953125000 (Pacific Daylight Time)

$FILE_NAME Attribute Values:
Flags: Archive
Name: m57biz.xls
Parent MFT Entry: 16176         Sequence: 3
Allocated Size: 294912          Actual Size: 291840
Created:        2008-07-19 18:28:03.687500000 (Pacific Daylight Time)
File Modified:  2008-07-19 18:28:03.953125000 (Pacific Daylight Time)
MFT Modified:   2008-07-19 18:28:03.953125000 (Pacific Daylight Time)
Accessed:       2008-07-19 18:28:03.953125000 (Pacific Daylight Time)

$OBJECT_ID Attribute Values:
Object Id: d9cb701e-55fa-11dd-ad9a-000c298d352b

Attributes:
Type: $STANDARD_INFORMATION (16-0)   Name: N/A   Resident   size: 72
Type: $FILE_NAME (48-4)   Name: N/A   Resident   size: 86
Type: $OBJECT_ID (64-5)   Name: N/A   Resident   size: 16
Type: $DATA (128-3)   Name: N/A   Non-Resident   size: 291840  init_size: 291840
```

9. What is the length of $DATA in bytes?

The length of the $DATA attribute is 291840 bytes, as identified using the 'istat' command.

```
Administrator in ~\Desktop\Evidence
istat -o 63 windows-image-01.dd 32712
MFT Entry Header Values:
Entry: 32712        Sequence: 2
$LogFile Sequence Number: 172434688
Allocated File
Links: 1

$STANDARD_INFORMATION Attribute Values:
Flags: Archive
Owner ID: 0
Security ID: 1067  (S-1-5-21-484763869-796845957-839522115-1004)
Created:        2008-07-19 18:28:03.656250000 (Pacific Daylight Time)
File Modified:  2008-07-19 18:28:03.953125000 (Pacific Daylight Time)
MFT Modified:   2008-07-19 18:28:04.046875000 (Pacific Daylight Time)
Accessed:       2008-07-19 18:28:03.953125000 (Pacific Daylight Time)

$FILE_NAME Attribute Values:
Flags: Archive
Name: m57biz.xls
Parent MFT Entry: 16176         Sequence: 3
Allocated Size: 294912          Actual Size: 291840
Created:        2008-07-19 18:28:03.687500000 (Pacific Daylight Time)
File Modified:  2008-07-19 18:28:03.953125000 (Pacific Daylight Time)
MFT Modified:   2008-07-19 18:28:03.953125000 (Pacific Daylight Time)
Accessed:       2008-07-19 18:28:03.953125000 (Pacific Daylight Time)

$OBJECT_ID Attribute Values:
Object Id: d9cb701e-55fa-11dd-ad9a-000c298d352b

Attributes:
Type: $STANDARD_INFORMATION (16-0)   Name: N/A   Resident   size: 72
Type: $FILE_NAME (48-4)   Name: N/A   Resident   size: 86
Type: $OBJECT_ID (64-5)   Name: N/A   Resident   size: 16
Type: $DATA (128-3)   Name: N/A   Non-Resident   size: 291840  init_size: 291840
```

Figure 29: TSK using 'istat' command to show metadata of $DATA

10. What is the allocated size for this file?

The allocated size for the file 'm57biz.xls' is 294,912 bytes, as identified using the 'istat' command.

```
Administrator in ~\Desktop\Evidence
istat -o 63 windows-image-01.dd 32712
MFT Entry Header Values:
Entry: 32712        Sequence: 2
$LogFile Sequence Number: 172434688
Allocated File
Links: 1

$STANDARD_INFORMATION Attribute Values:
Flags: Archive
Owner ID: 0
Security ID: 1067  (S-1-5-21-484763869-796845957-839522115-1004)
Created:        2008-07-19 18:28:03.656250000 (Pacific Daylight Time)
File Modified:  2008-07-19 18:28:03.953125000 (Pacific Daylight Time)
MFT Modified:   2008-07-19 18:28:04.046875000 (Pacific Daylight Time)
Accessed:       2008-07-19 18:28:03.953125000 (Pacific Daylight Time)

$FILE_NAME Attribute Values:
Flags: Archive
Name: m57biz.xls
Parent MFT Entry: 16176         Sequence: 3
Allocated Size: 294912          Actual Size: 291840
Created:        2008-07-19 18:28:03.687500000 (Pacific Daylight Time)
File Modified:  2008-07-19 18:28:03.953125000 (Pacific Daylight Time)
MFT Modified:   2008-07-19 18:28:03.953125000 (Pacific Daylight Time)
Accessed:       2008-07-19 18:28:03.953125000 (Pacific Daylight Time)
```

Figure 30: TSK using 'istat' command to show allocated size

11. What is the actual size for this file?

The actual size of the file ‘m57biz.xls’ is 291,840 bytes, as identified using the ‘istat’ command.

```
Administrator in ~\Desktop\Evidence
istat -o 63 windows-image-01.dd 32712
MFT Entry Header Values:
Entry: 32712        Sequence: 2
$LogFile Sequence Number: 172434688
Allocated File
Links: 1

$STANDARD_INFORMATION Attribute Values:
Flags: Archive
Owner ID: 0
Security ID: 1067  (S-1-5-21-484763869-796845957-839522115-1004)
Created:        2008-07-19 18:28:03.656250000 (Pacific Daylight Time)
File Modified:  2008-07-19 18:28:03.953125000 (Pacific Daylight Time)
MFT Modified:   2008-07-19 18:28:04.046875000 (Pacific Daylight Time)
Accessed:       2008-07-19 18:28:03.953125000 (Pacific Daylight Time)

$FILE_NAME Attribute Values:
Flags: Archive
Name: m57biz.xls
Parent MFT Entry: 16176         Sequence: 3
Allocated Size: 294912          Actual Size: 291840
Created:        2008-07-19 18:28:03.687500000 (Pacific Daylight Time)
File Modified:  2008-07-19 18:28:03.953125000 (Pacific Daylight Time)
MFT Modified:   2008-07-19 18:28:03.953125000 (Pacific Daylight Time)
Accessed:       2008-07-19 18:28:03.953125000 (Pacific Daylight Time)
```

Figure 31: TSK using ‘istat’ command to show actual size

12. What is the file slack for this file?

The file slack (used space) is 3,072 bytes, calculated as the difference between the allocated size (294,912 bytes) and the actual size (291,840 bytes).

```
Administrator in ~\Desktop\Evidence
istat -o 63 windows-image-01.dd 32712
MFT Entry Header Values:
Entry: 32712        Sequence: 2
$LogFile Sequence Number: 172434688
Allocated File
Links: 1

$STANDARD_INFORMATION Attribute Values:
Flags: Archive
Owner ID: 0
Security ID: 1067  (S-1-5-21-484763869-796845957-839522115-1004)
Created:        2008-07-19 18:28:03.656250000 (Pacific Daylight Time)
File Modified:  2008-07-19 18:28:03.953125000 (Pacific Daylight Time)
MFT Modified:   2008-07-19 18:28:04.046875000 (Pacific Daylight Time)
Accessed:       2008-07-19 18:28:03.953125000 (Pacific Daylight Time)

$FILE_NAME Attribute Values:
Flags: Archive
Name: m57biz.xls
Parent MFT Entry: 16176         Sequence: 3
Allocated Size: 294912          Actual Size: 291840
Created:        2008-07-19 18:28:03.687500000 (Pacific Daylight Time)
File Modified:  2008-07-19 18:28:03.953125000 (Pacific Daylight Time)
MFT Modified:   2008-07-19 18:28:03.953125000 (Pacific Daylight Time)
Accessed:       2008-07-19 18:28:03.953125000 (Pacific Daylight Time)
```

Figure 32: TSK using 'istat' command to find slack size

13. **Using fls, what is the metadata reference for the folder where this file resides?**

Using the ‘fls’ command, the metadata reference for the folder containing the file ‘m57biz.xls’ was identified as 16176, which corresponds to the Desktop directory within the Jean user profile. This was confirmed by correlating the file path obtained using the ‘ffind’ command with the directory listings from ‘fls’.

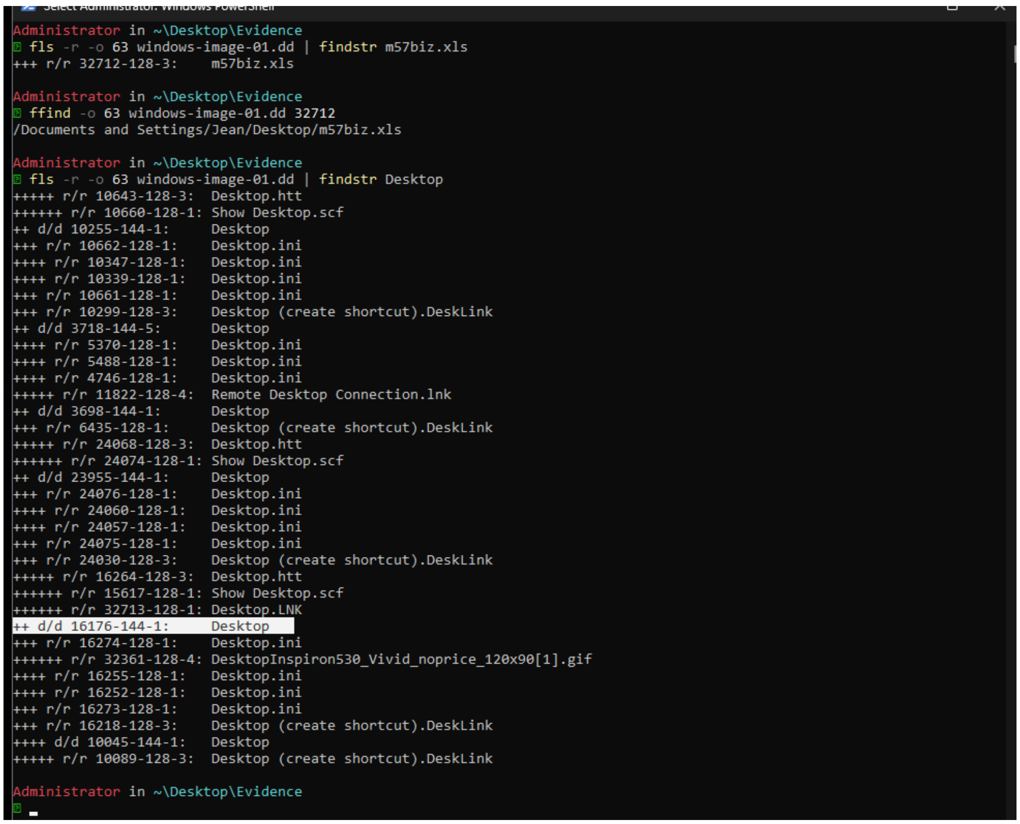


Figure 33: TSK using 'fls' command to find metadata reference for Desktop folder

14. **Generate a bodyfile for this metadata reference only. How many entries are their?**

A bodyfile was generated for the directory (inode 16176) using the ‘fls’ command. The bodyfile contains 4 entries, representing files and directories within the folder.

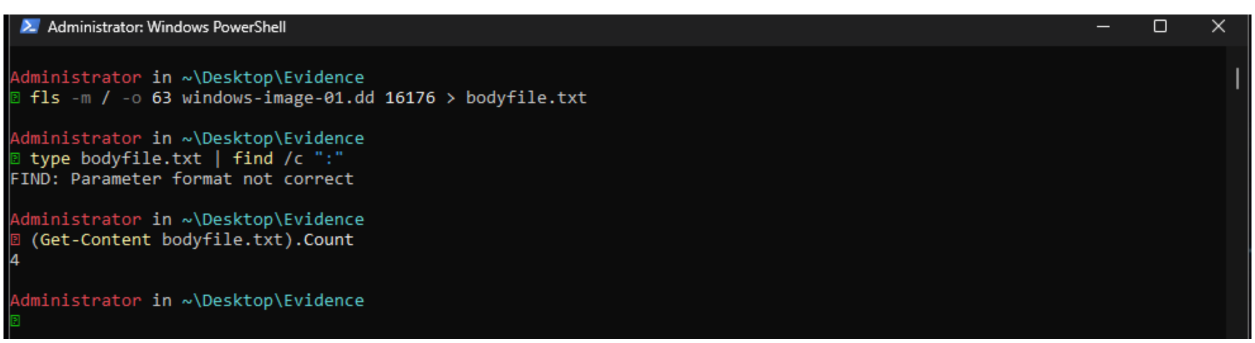


Figure 34: Using bodyfile to count how many entries there are

15. **Generate a timeline using mactime for this bodyfile. What is the modified, accessed, created (born) date timestamp for the targetfile? (Hint: ma.b)**

A forensic timeline was constructed based on file system metadata obtained using the 'istat' command. The timestamps for the file 'm57biz.xls' are as follows:

- Modified: 2008-07-19 18:28:03.953125000 (Pacific Daylight Time)
- Accessed: 2008-07-19 18:28:03.953125000 (Pacific Daylight Time)
- Created (Born): 2008-07-19 18:28:03.656250000 (Pacific Daylight Time)

```
$STANDARD_INFORMATION Attribute Values:
Flags: Archive
Owner ID: 0
Security ID: 1067  (S-1-5-21-484763869-796845957-839522115-1004)
Created:        2008-07-19 18:28:03.656250000 (Pacific Daylight Time)
File Modified:  2008-07-19 18:28:03.953125000 (Pacific Daylight Time)
MFT Modified:   2008-07-19 18:28:04.046875000 (Pacific Daylight Time)
Accessed:       2008-07-19 18:28:03.953125000 (Pacific Daylight Time)
```

16. **What is the MFT changed date timestamp for the target file? (Hint: ..c.)**

The MFT changed timestamp for the file 'm57biz.xls' is **2008-07-19 18:28:04.046875000 (Pacific Daylight Timeline)**, as identified using the 'istat' command.


```
$STANDARD_INFORMATION Attribute Values:
Flags: Archive
Owner ID: 0
Security ID: 1067  (S-1-5-21-484763869-796845957-839522115-1004)
Created:        2008-07-19 18:28:03.656250000 (Pacific Daylight Time)
File Modified:  2008-07-19 18:28:03.953125000 (Pacific Daylight Time)
MFT Modified:   2008-07-19 18:28:04.046875000 (Pacific Daylight Time)
Accessed:       2008-07-19 18:28:03.953125000 (Pacific Daylight Time)
```

Conclusion

This assignment provided valuable experience in performing both low-level disk analysis and advanced file system analysis using WinHex and The Sleuth Kit. It strengthened my understanding of partition structures, NTFS metadata, and how to identify and verify specific files within a forensic image. The additional tasks of analyzing MFT attributes, calculating file slack, and constructing a forensic timeline enhanced my ability to interpret file activity and understand how user interactions are recorded within the file system. Overall, this assignment demonstrated a realistic forensic workflow and emphasized the importance of accuracy, attention to detail, and clear documentation when investigating potential data exfiltration incidents.

References

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